Determine Equivalent Ratios Using Graphs

Read what each point on a ratio graph means.

Name _____ **Date** _____

Tonight's spark

A farm stand graph shows bags across and apples up. One dot sits at (3, 18). The first number is bags. How many apples are in just one bag?

Make a guess first: does one bag hold more than 10 apples, or fewer?

Boxes (x)	Crayons (y)
0	0
1	9
4	36
7	?

×7

Our class model: the unit rate point

Try it

Work it out on paper. Write one answer on each line.

1. **The line for minutes (x) and pages (y) passes through (1, 7). How many pages are read in 1 minute?**

   A. 1 **B.** 8 **C.** 7 **D.** 6

   My answer _____

2. **The point (5, 30) is on a graph of bags (x) and marbles (y). What does it mean?**

   A. 5 marbles are in 30 bags **B.** 5 bags hold 30 marbles **C.** 30 bags hold 5 marbles
   D. 5 + 30 = 35 marbles

   My answer _____

3 **The unit rate is 9 miles per hour. Hours are x and miles are y. Which point is on the line?**

A. (9, 1) **B.** (10, 1) **C.** (1, 8) **D.** (1, 9)

My answer _____

4 **The unit rate is 6. Which point is on the graph?**

A. (4, 24) **B.** (4, 10) **C.** (24, 4) **D.** (6, 4)

My answer _____

5 **On a graph of hours (x) and miles (y), what does the point (0, 0) mean?**

A. 0 hours go with 1 mile **B.** 1 hour goes with 0 miles **C.** 0 hours go with 0 miles

My answer _____

6 **A line passes through (1, 5). Use the unit rate to find y when x is 8.**

A. 13 **B.** 40 **C.** 3 **D.** 5

My answer _____

Talk about it

What does the point with x = 1 tell you about this rate?

Home mission

Pretend each bag holds 6 apples. Ask your student to say the point for 4 bags as a sentence. Then ask: what does the point (0, 0) mean?

If we get stuck, ask

- Which quantity is on the x-axis? Read that number first.
- What does the point where x is 1 tell us?
- Can you say this point as a sentence about real things?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/3-4-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it